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Next Generation AI Researchers for Air Quality Excellence



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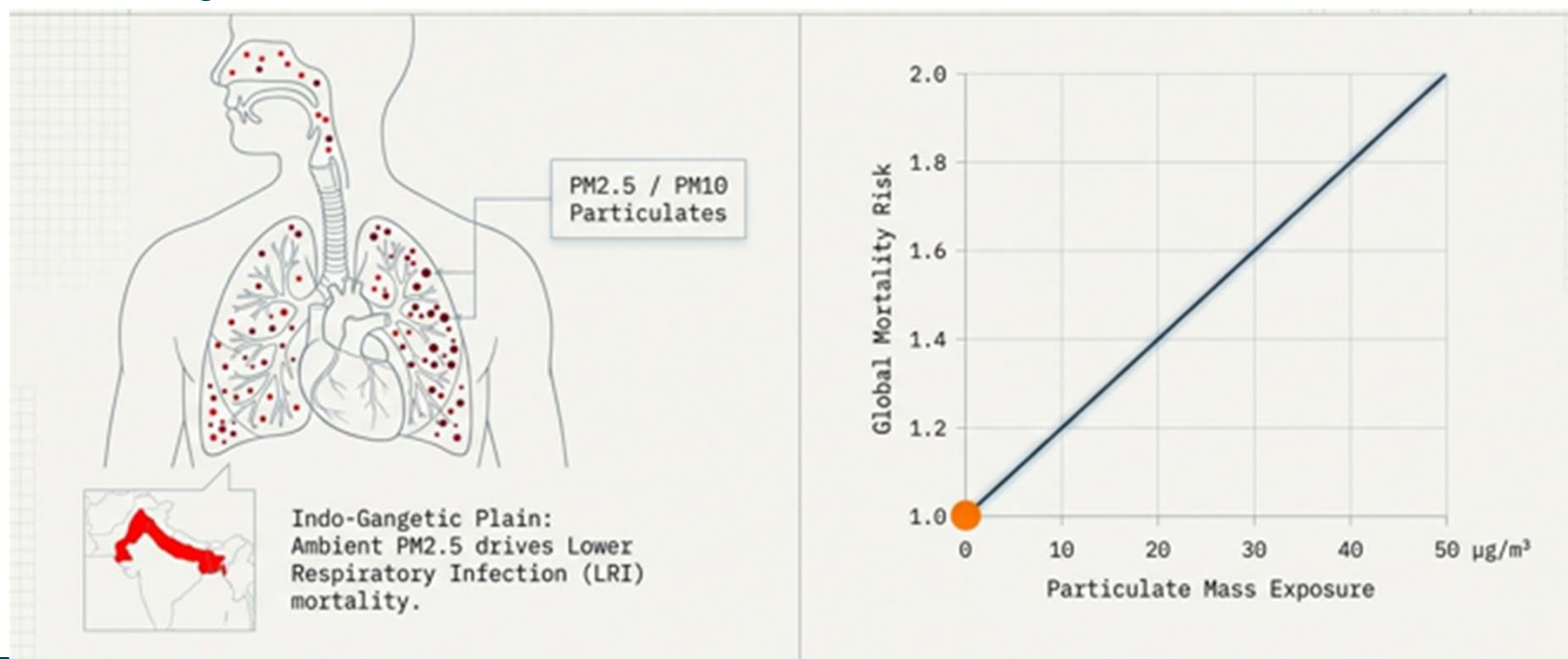
Environmental Informatics & Spatial Architecture



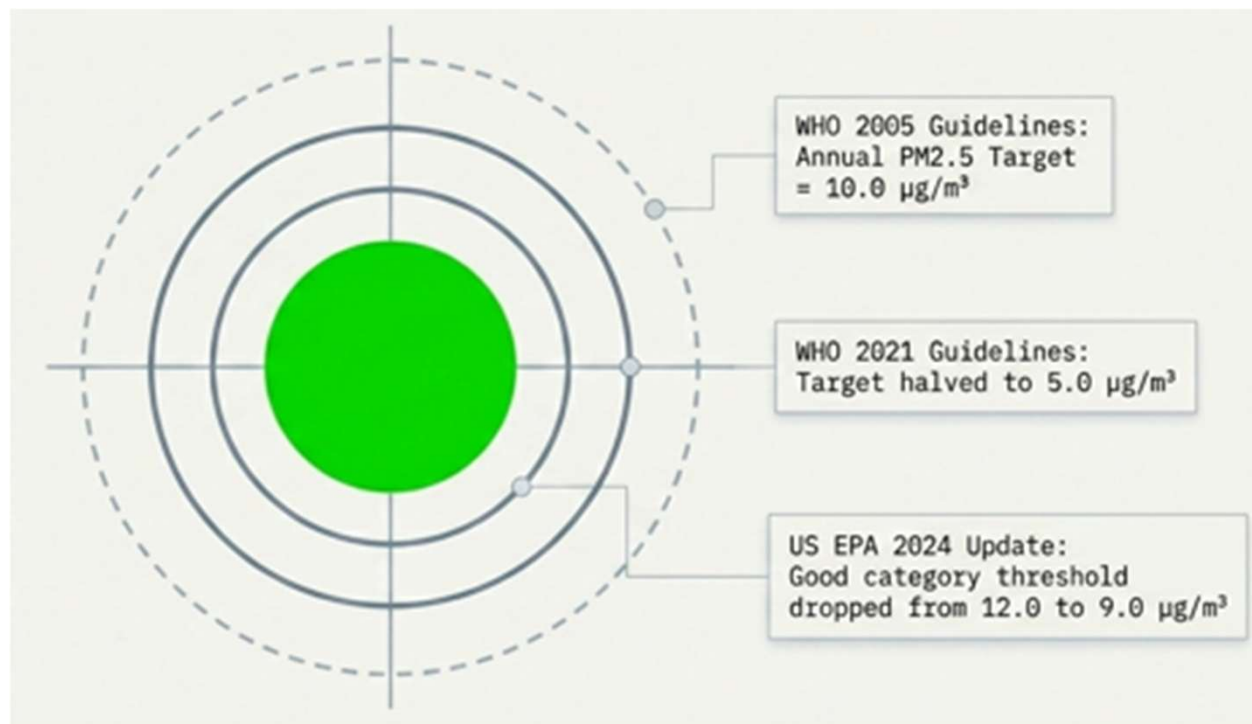
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The epidemiological paradigm

- Ambient particulate mass exposure is a primary driver of global mortality. The linear exposure-response curve demonstrates an absence of any safe threshold, dictating the need for continuous, highly localized programmatic tracking.



The shrinking safe zone



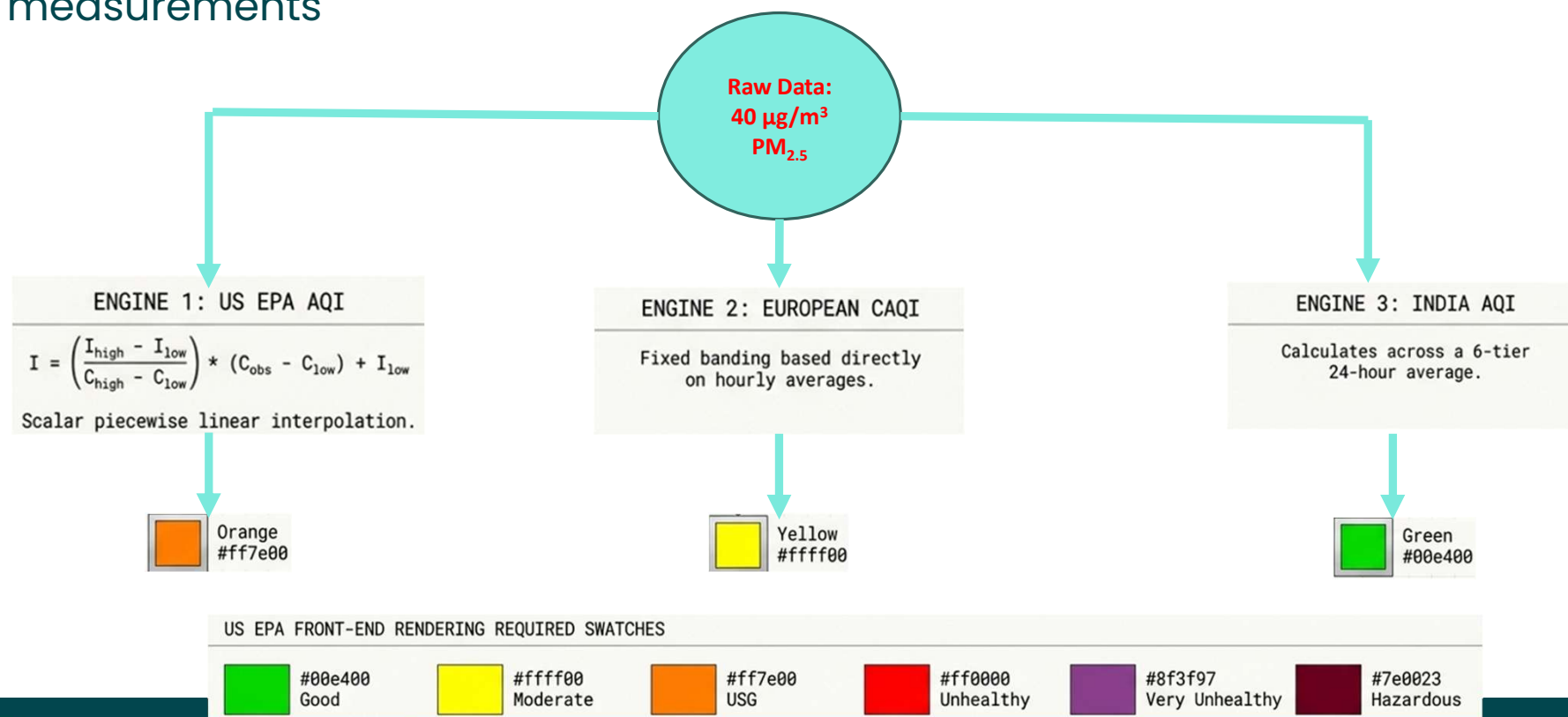
A raw reading of 10.5 $\mu\text{g}/\text{m}^3$ used to return Code Green. As of 2024, the exact same reading must trigger Code Yellow.



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The math behind the map

- Air Quality Indices are localized mathematical mappings, not universal measurements



Data ingestion architecture

OpenAQ (v3 RESTful)

Use Case:

Best for scientific parity and historical analysis.

Data Output:

Raw physical concentrations ($\mu\text{g}/\text{m}^3$ or particles/ cm^3).

Key Endpoints:

/v3/locations

Parameter ID 2 (PM2.5), ID 1 (PM10)

Note:

v1 and v2 endpoints retired January 31, 2025.

WAQI (World Air Quality Index)

Use Case:

Best for real-time public alerts and live map rendering.

Data Output:

Pre-calculated localized AQI scraped from global EPAs.

Key Endpoints:

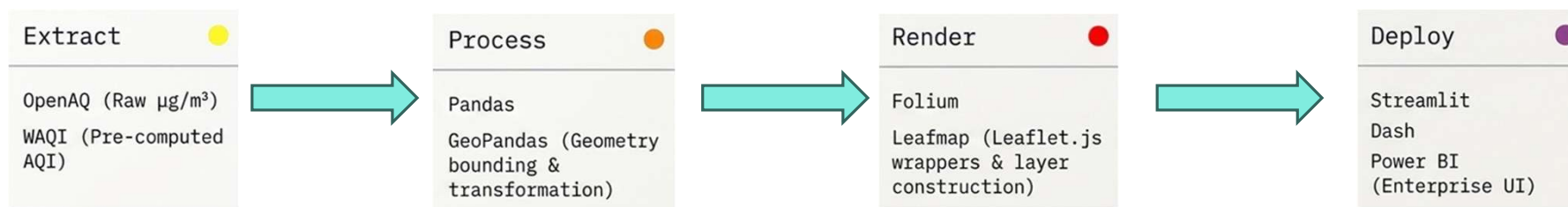
/feed/

Bounding box queries for lightweight payloads.

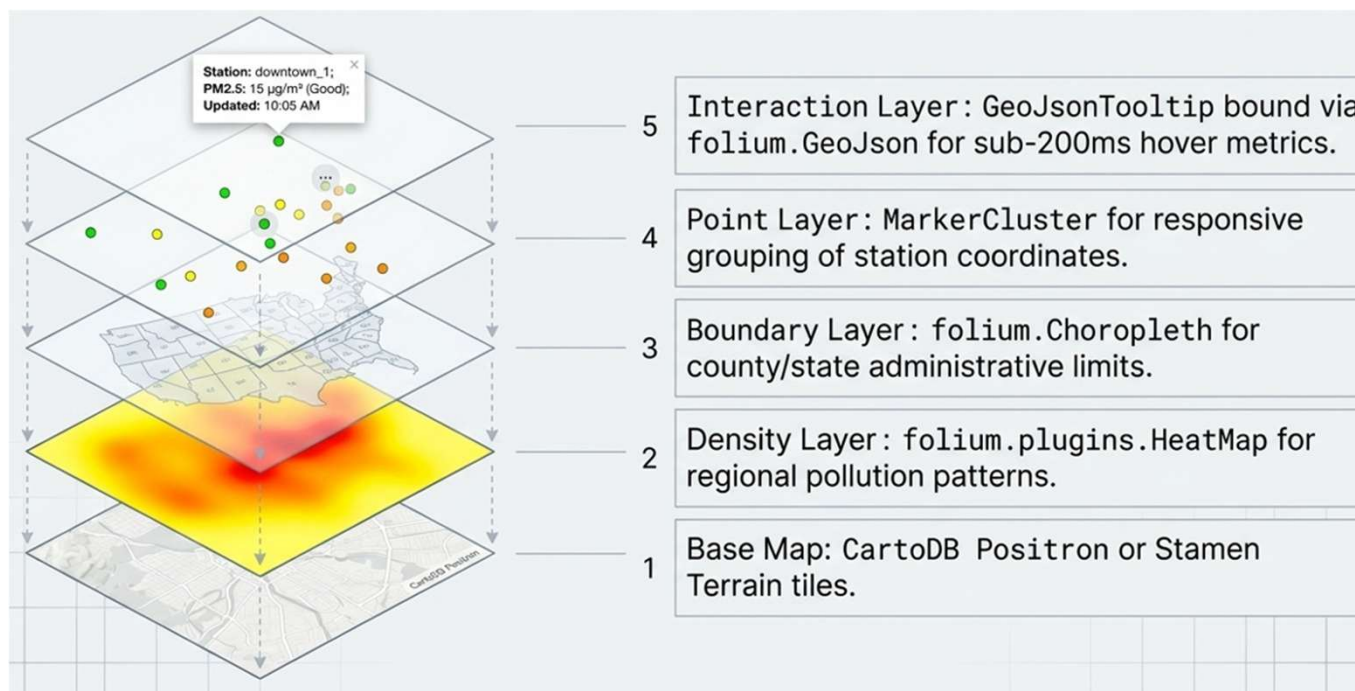


The modern spatial stack

- End-to-End Architecture: from Pipeline to Dashboard



Map Construction: The Isometric Layer Stack



Dashboard deployment frameworks

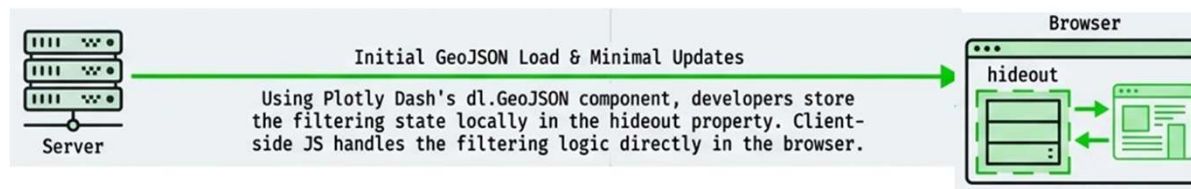
Streamlit (streamlit-folium)	Dash (dash-leaflet)	NiceGUI (ui.html)
State: Bidirectional via <code>st_folium()</code>. Constraint: Re-runs entire script on interaction. Requires <code>@st.cache_data</code> to prevent full-page reload lag.	State: Virtual DOM-based React wrapper. Advantage: High limits for client-side rendering; excellent for dense vector layers.	State: Unidirectional iframe wrapper via <code>_repr_html_()</code>. Constraint: Relies on custom JS injection to communicate actions outside the iframe.

Optimizing State: the dash-leaflet hideout method

- The Bottleneck (Failing Server – Side Flow)
 - This approach involves sending the entire GeoJSON dataset to the server for filtering on each user interaction and then receiving the update dataset back, leading to significant performance degradation and poor user experience.

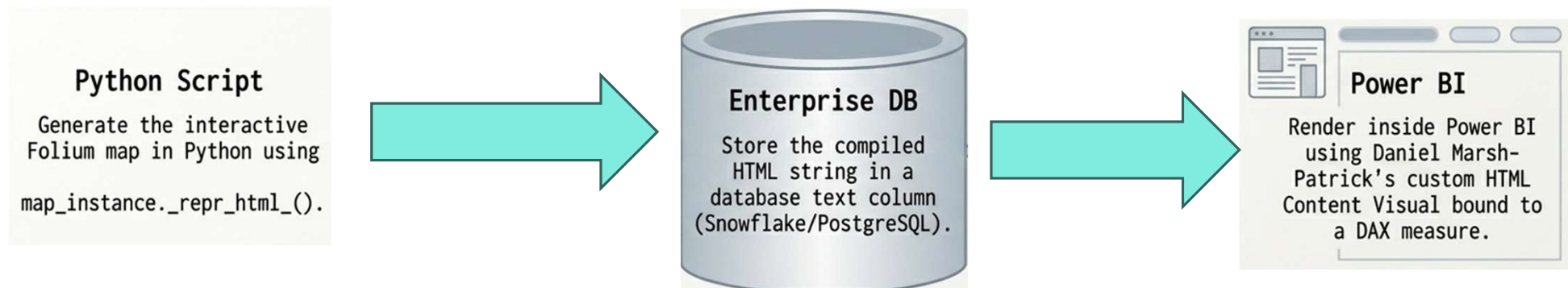


- The Solution (Optimized Client-Side Flow)
 - By storing the filtering state within the 'hideout' property of the 'dl.GeoJSON' component, the data processing is offloaded to the client's browser, significantly reducing server load and resulting in instant, lag-free UI interactions.



Enterprise BI integration (The Power BI Challenge)

- Moving custom Python maps into restrictive enterprise environments



- THE CHALLENGE: Native Power BI mapping tools cannot render complex spatial layers like multi-pollutant density heatmaps or custom clusters

Navigating Sandbox Security

- Bypassing Power BI's iframe security restrictions.



Problem 1:
CORS & Domain
Isolation Blocks



Fix: Point map tiles to permissive
self-hosted tile servers using
`Access-Control-Allow-Origin: *`.



Problem 2:
Local File Blocks



Fix: Compile self-contained HTML payloads
directly in Python (inline all CSS/JS).



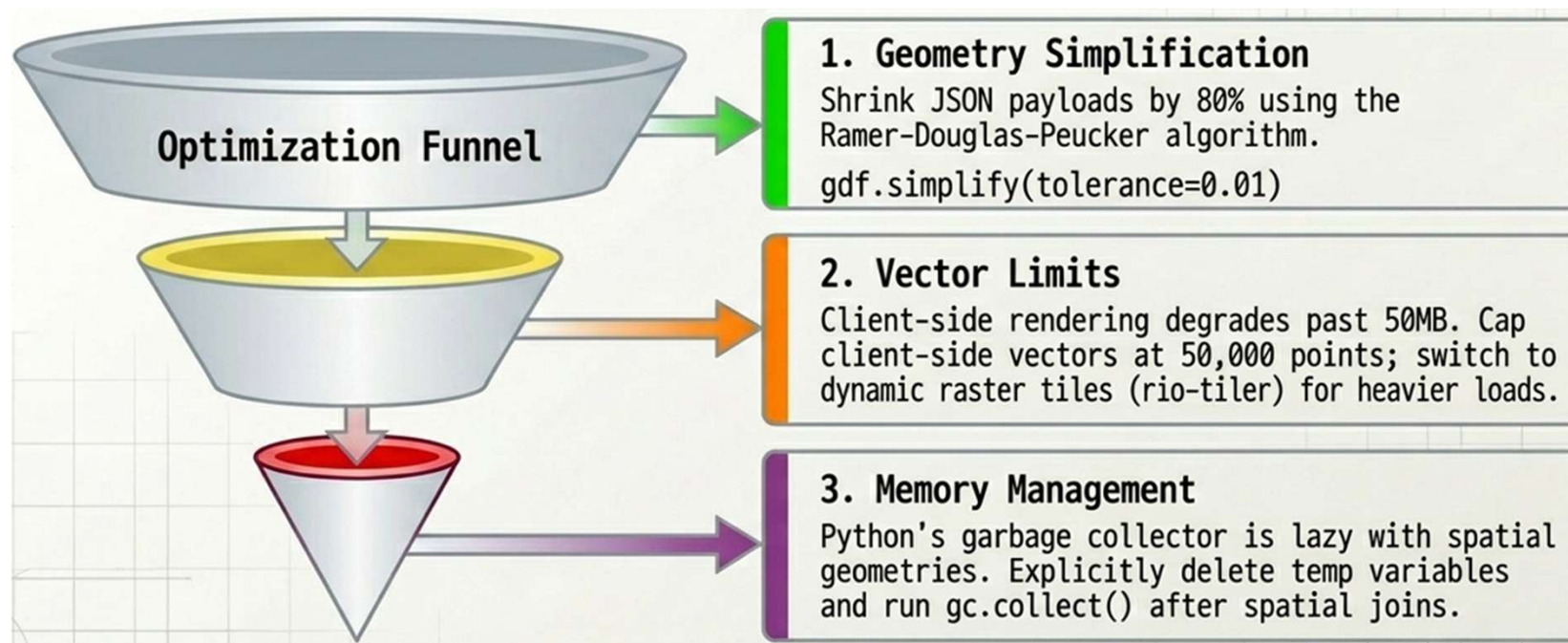
Problem 3:
Strict Corporate
Network Blocks



Fix: Deploy the Certified 'HTML Content
(lite)' visual and generate entirely
self-contained SVG vectors via DAX to
eliminate all external network requests.

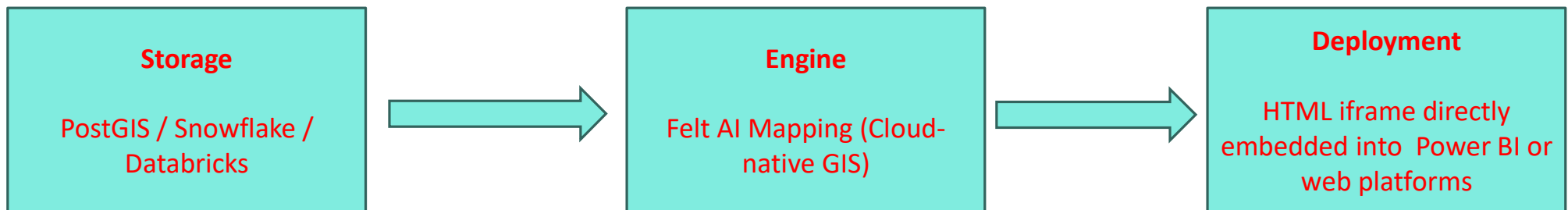
Scaling the System (Performance Thresholds)

- Edge performance and optimization for massive geospatial datasets



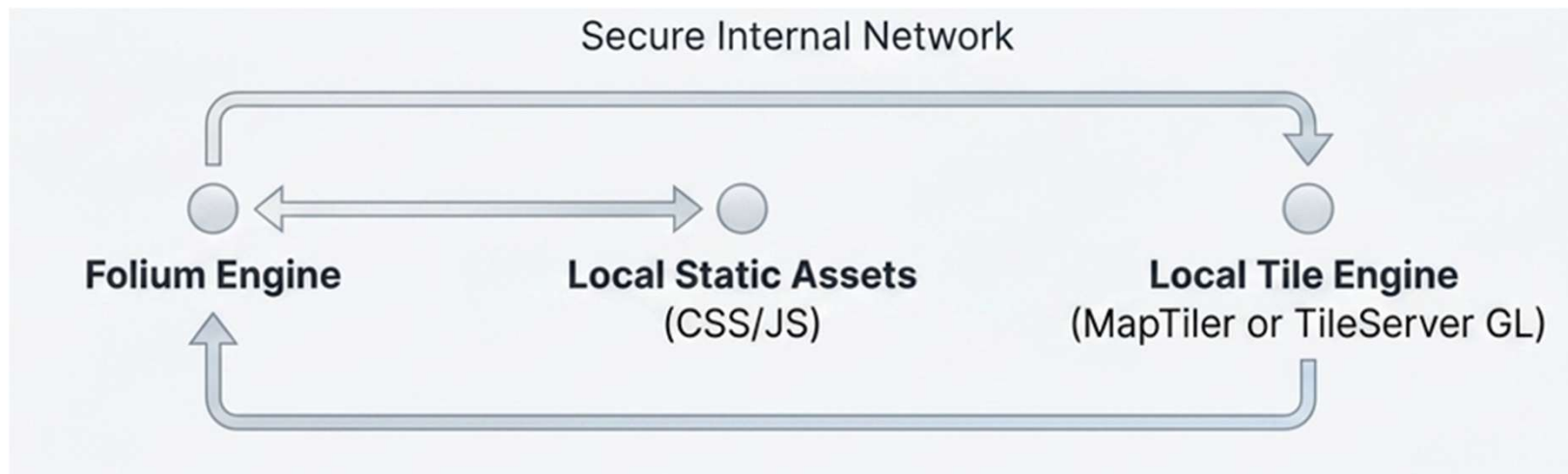
Cloud-Native Alternatives: Felt & Databricks

- Streamlined cloud-native workflow for high-performance spatial intelligence
- Key advantage: Bypassing local Python rendering eliminates the need for local geo-processing overhead while maintaining high-performance spatial intelligence.



Air-Gapped Deployments

- Running spatial architecture securely without external internet access
- Override Folium's default configurations to load Leaflet assets from local static folders
- Deploy a local tile engine (MapTiler Server or TileServer GL)
- Point the map's tiles parameter strictly to the internal server: `http://internal-tiles/{z}/{x}/{y}.png`.



The Synthesis Blueprint

- Summary of Environmental BI Architecture

Q1: Data Truth (Extract) Always ingest and store raw physical concentrations ($\mu\text{g}/\text{m}^3$). Compute localized AQI dynamically based on the latest EPA/WHO breakpoints.	Q2: Geoprocessing (Transform) Enforce RDP geometry simplification. Utilize columnar Parquet files and DuckDB for heavy querying.
Q3: Presentation (UI) Offload filtering to the client-side browser using Dash hideout to ensure sub-200ms interaction speeds.	Q4: Enterprise (Deploy) Respect the Power BI sandbox. Rely on inline HTML, DAX-generated SVGs, or Azure Service Principals for secure embedding.



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Thank You!



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